Ádám Fodor

PhD Research Scientist

Contact

+36-30-993-2313

fodorad201@gmail.com

adamfodor.com

linkedin.com/in/fodorad

github.com/fodorad

Class B driving license

Budapest, Hungary

About Me

PhD candidate with over 11 years of research and development experience in machine learning and computer vision. Focused on Human-Centered AI for personality research and affective computing. Experienced lecturer and supervisor.

Languages

- Hungarian – native
- English – fluent

Skills

- Data analysis (pandas, NumPy)
- Deep Learning (PyTorch, scikit-learn)
- Large Language Models (transformer architectures)
- Computer vision & multimodal AI
- Experimental design, model evaluation, statistical analysis, reproducibility practices
- Cross-platform development
- Programming languages (Python, Bash scripting)
- Version control (Git, GitHub Actions, Travis CI/CD)
- Project management (Trello, MS Teams, Slack)
- Problem solving & analytical thinking
- Strong independent developer
- Proactive team collaborator
- Effective communication (presenting research findings, technical writing and reporting)

Awards

- 1st place Students' Scientific Association Conference (TDK), 2018
- Special award, Morgan Stanley, 2018
- Student TDK supervision award, 2023

Experience

Research Scientist — since 09/2017

Neural Information Processing Group (NIPG)
ELTE-IK, Department of Artificial Intelligence

- Worked on *video processing and object tracking* as part of project EFOP-3.6.3-VEKOP-2017-00002, resulting in **two rank A conference** publications at IJCNN.
- Developed an *AI-based eye blink detection pipeline* for non-verbal communication analysis, published in a **Q2 journal**, and later integrated into a composite AI framework for automatic behavior analysis of children with autism spectrum disorder, resulting in a subsequent **Q1 journal** publication.
- Integrated *3D avatars with large language models* to create realistic virtual agents capable of gesture-based communication in augmented reality.
- Partnered with **Barcelona University** on *multimodal sentiment and personality perception research*, leading to publication and strengthening NIPG partnerships.
- Collaborated with the **German Research Center for Artificial Intelligence (DFKI)** in a 4-member team on *behavior and gaze analysis*, enhancing NIPG's reputation.
- Contributed as part of a 5-member team with **ELTE Faculty of Education and Psychology** on infant–mother communication analysis, deepening NIPG partnerships, and resulting in 2 BSc and 2 MSc theses.
- Conducted a *speech de-identification* project as part of my MSc thesis, which earned **TDK awards** and led to publication.
- *Teaching MSc courses* on the mathematical foundations and applications of state-of-the-art AI (Large Language Models included), *supervising* students (BSc and MSc), and guiding a MSc student's TDK project.
- Managed 40+ research databases supporting 10+ NIPG projects over the past 3 years.

Associate Scientist — 07/2014 – 08/2017

Institute for Computer Science and Control
Hungarian Academy of Sciences

- **Executed a basic research project** for OPEL Szentgotthárd Kft. as part of iKOMP, applying feature selection and production trend forecasting methods to *reduce engine failures*.
- Developed a *novel feature selection technique* called AHFS, published in a **Q1 journal**, and successfully *applied in real-world tasks* such as situation detection in specialized machining and wind turbine bearing temperature monitoring.
- Designed a *software framework for AI model training and data visualization* as part of my BSc thesis, extending prior tools to support data mining; created in collaboration with MTA SZTAKI and *adopted as a successor to their in-house system*.

Selected Publications

- Bruno Carlos Dos Santos Melício, Kaan Karaköse, **Ádám Fodor**, Linyun Xiang, Viktor Varga, Latha Soorya, Emily Dillon, Péter Kun, András Sárkány, Mohamed Chetouani, Kristian Fenech; *Multimodal Framework for Automatic Behavior Analysis of Children with Autism During ADOS-2, Cognitive Computation*, vol. 17, 137, 2025
- **Ádám Fodor**, Kristian Fenech, András Lőrincz; *BlinkLinMulT: Transformer-Based Eye Blink Detection*, MDPI Journal of Imaging, vol. 9-10, 196, 2023
- Zsolt János Viharos, Krisztián Balázs Kis, **Ádám Fodor**, Máté István Büki; *Adaptive, hybrid feature selection (AHFS)*, Pattern Recognition, vol. 116, 107932, 2021
- [For further 7 publications, please visit my website.]

Education

PhD candidate in Computer Science — 08/2019 – 08/2025

Eötvös Loránd University (ELTE)

- Thesis: Affective Behavior and Social Interaction Analysis
- Supervisor: Dr. habil. András Lőrincz
- Submitted, successful departmental defense, public defense is upcoming

MSc in Computer Science — 01/2016 – 07/2018

Eötvös Loránd University (ELTE)

- Thesis: Speech de-identification with deep neural networks
- Supervisor: Dr. habil. András Lőrincz
- Qualification: excellent

References

Dr. Kristian Fenech
Head of NIPG, ELTE
Email : fenech@inf.elte.hu

Dr. habil. András Lőrincz
Founder of NIPG, ELTE
Email : lorincz@inf.elte.hu